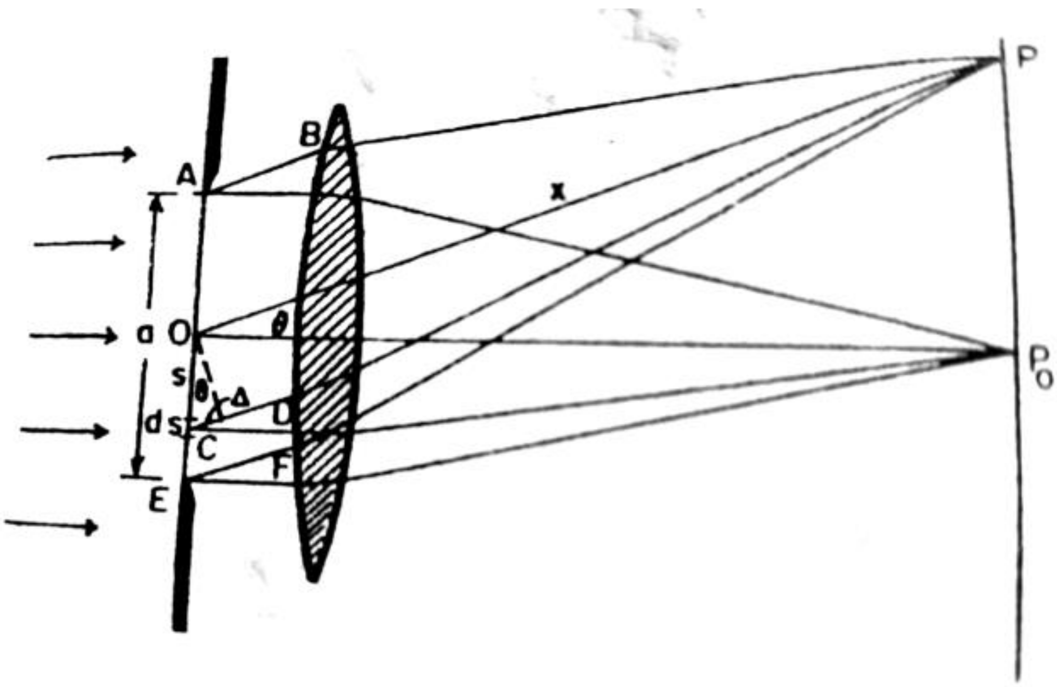

Figure 1

Fraunhofer Diffraction (Single Slit):

The wave equation of light at the slit AE at a distance r from the source may be written as

$$y = \frac{A}{r}\sin\left(\omega t - kr\right)$$

Where A is the amplitude at a unit distance from the source.
Let a be the width of the slit and ds be an element of the width of the wavefront in the plane of the slit, at a distance s from O, as shown in Figure 1.
The wavelet produced by the element ds makes a displacement at $P_{0,}$ which is written as

$$dy = \frac{rds}{x_0}\sin\left(\omega t - kx_0\right)$$

Where x_0 is the distance from s to P_o.
Now if the wavelet from O covers a distance x to reach the screen at P, the wave equation becomes

$$dy_0 = \frac{rds}{x}\sin\left(\omega t - kx\right)$$

The part of the wavelet from ds that reaches P covers an extra Δx path. Thus, we may write

$$dy_s = \frac{rds}{x}\sin\{\omega t - k(x + \Delta x)\}$$

or,$dy_s = \frac{rds}{x}\sin(\omega t - kx - k\Delta x)$

or, $dy_s = \frac{rds}{x}\sin(\omega t - kx - kssin\theta)$

The contributions from all the wavelets produced between A and E to the location P at the screen may be found by integrating

$$y = \int_{-a/2}^{a/2} dy_s$$

or, $y = \frac{r}{x}\int_{-a/2}^{a/2}\sin\{(\omega t - kx) - kssin\theta\}\,ds$

$$\text{or,}y = \frac{r}{x}\int_{-a/2}^{a/2}[\sin(\omega t - kx)\cos(kssin\theta) - \cos(\omega t - kx)\sin(kssin\theta)\, ds$$

$$\text{or,}y = \frac{r}{x}\left[\sin(\omega t - kx)\frac{\sin(kssin\theta)}{ksin\theta} + \cos(\omega t - kx)\frac{\cos(kssin\theta)}{ksin\theta}\right]_{-a/2}^{a/2}$$

$$\text{or,}y = \frac{r}{x}\left[\sin(\omega t - kx)\frac{\sin\left(k\frac{a}{2}sin\theta\right)}{ksin\theta} + \cos(\omega t - kx)\frac{\cos(k\frac{a}{2}sin\theta)}{ksin\theta} - \sin(\omega t - kx)\frac{\sin\left(-k\frac{a}{2}sin\theta\right)}{ksin\theta} - \right.$$

$$\left.\cos(\omega t - kx)\frac{\cos(-k\frac{a}{2}sin\theta)}{ksin\theta}\right]$$

$$\text{or, } y = \frac{r}{x}\frac{2sin\left(k\frac{a}{2}sin\theta\right)}{ksin\theta}sin(\omega t - kx)$$

$$\text{or,}y = \frac{ra}{x}\frac{sin\left(k\frac{a}{2}sin\theta\right)}{k\frac{a}{2}sin\theta}sin(\omega t - kx)$$

$$y = R_0\frac{sin\beta}{\beta}\ sin(\omega t - kx)$$

where R_0 = ra/x and $\beta = k\frac{a}{2}sin\theta$

We know that,

Intensity = $(\text{Amplitude})^2$

$$\text{Thus, } I = R_0^2\frac{sin^2\beta}{\beta^2}$$

Position of principle/central maximum:

$$\text{For maximum intensity, } I = R_0^2\text{, we have } \lim_{\beta\to 0}\frac{sin\beta}{\beta} = 1\text{,which implies } \beta = 0$$

$$\text{or, } k\frac{a}{2}sin\theta = 0$$

$$\text{or,}sin\theta = 0$$

$$\text{or,}\theta = 0$$

Position of minimum intensity:

I=0 if and only if $sin\beta = 0$ which implies $\beta = m\pi$ where $m = \pm 1, \pm 2, \pm 3$, ….

$$k\frac{a}{2}sin\theta = m\pi$$

$$\text{or, } \frac{2\pi}{\lambda}\frac{a}{2}sin\theta = m\pi$$

$$\text{or, } asin\theta = m\lambda$$

Position of secondary maxima:

$$I = R_0^2\frac{sin^2\beta}{\beta^2}$$

The condition for finding the maxima or minima is given by

$$\frac{dI}{d\beta} = 0$$

$$\text{or}R_0^2\frac{\beta^2 2sin\beta cos\beta - sin^2\beta 2\beta}{\beta^4} = 0$$

$$\text{or, } sin\beta cos\beta(\beta - tan\beta) = 0$$

$$\text{or, } sin\beta(\beta - tan\beta) = 0\text{, because } cos\beta \neq 0$$

Either $sin\beta = 0$ or $\beta - tan\beta = 0$

$sin\beta = 0$, gives the position of minima except $\beta = 0$ which corresponds to the central maxima.

For the other maxima, we have

$$\beta - tan\beta = 0$$

$$\text{or, } \beta = tan\beta$$

Which is a transcendental equation. Solving it from the graph as shown in Figure 2, we get

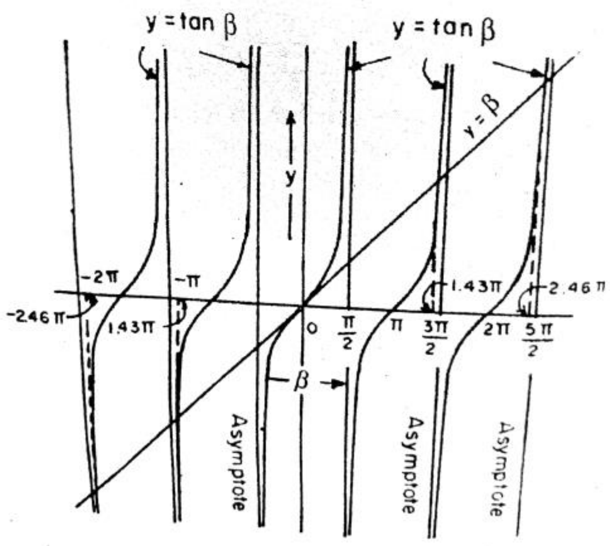

Figure 2

$$\beta = \pm 1.43\pi, \pm 2.46\pi, \pm 3.47, \pm 4.48, \ldots$$

Intensity pattern:

The positions of maxima obtained for

$$\beta = 0, \pm 1.43\pi, \pm 2.46\pi, \pm 3.47\pi, \pm 4.48\pi, \ldots$$

The positions of minima obtained for

$$\beta = \pm \pi, \pm 2\pi, \pm 3\pi, \pm 4\pi, \ldots$$

The intensity pattern graph is shown in Figure 3.

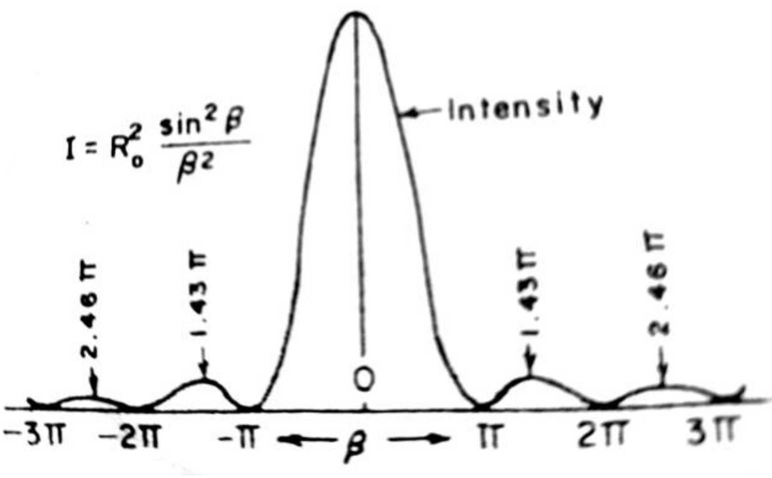

Figure 3